

TABLE 1: A COMPARISON OF LOW EARTH ORBIT (LEO), MEDIUM EARTH ORBIT (MEO), AND GEOSTATIONARY EARTH ORBIT (GEO) SATELLITES

	LEO	MEO	GEO
Altitude	200-2000km	2000-35,786km	35,786km
Orbital period	90-120 minutes	2-24 hours	24 hours
Latency (round trip)	5-20 milliseconds (ms)	100-180ms	500-700 + ms
Coverage area	Small (moves quickly)	Medium	Large (1/3rd of Earth)
Satellites required for global coverage	A constellation of thousands of satellites	10 to 30 satellites	Three strategically placed satellites
Visibility window	10-15 minutes	2-8 hours	Continuous
Power requirements	Low	Medium	High
Launch cost	Low	Medium	High
Signal strength	Strong	Medium	Weak
Tracking complexity	High (fast-moving)	Medium	None (stationary)
Lifespan	5-10 years	10-15 years	15+ years
Typical applications	Internet (Starlink, OneWeb), Earth observation, remote sensing, International Space Station (ISS)	Navigation (GPS, Galileo, GLO-NASS), communications (O3b)	TV/radio broadcasting, weather monitoring, satellite communication for static locations
Examples	Starlink, OneWeb, Iridium	GPS, Galileo, GLONASS	INSAT, GOES

Note: All figures are approximate; Source: Claude Opus 4 and Gemini 2.5 Pro on Yupp.ai

centres within India to ensure data localisation, service auditability, and monitoring. This is very important because satellite communications are being used not only by the government and military of different nations but also by the mafia, drug traffickers, terrorists, and other anti-social elements for clandestine communications. Hence, it is important to be able to track traffic, localise user data, and enforce shutdowns when necessary.

Airtel, in collaboration with OneWeb, and Jio, in partnership with SES, have also demonstrated successful proof-of-concepts in certain regions, with provisional spectrum allocation. Meanwhile, Starlink and Amazon are planning a hybrid approach, where they will offer their services both directly and through partners. They have started inking pacts with existing satcom players, such as Hughes Communications, Nelco, and Inmarsat, to quickly enter the business-to-business (B2B) and business-to-government (B2G) markets, and with telecom majors like Airtel and Jio to

tap the business-to-consumer (B2C) market.

The telecom industry is now eagerly waiting for the final nod—the allocation of spectrum for satellite communications. The Telecom Regulatory Authority of India (TRAI) has recommended that the DoT should allocate the spectrum administratively instead of holding an auction to speed up the launch of satellite internet services. The model would require satellite companies to pay a minimum annual charge per megahertz (MHz) and ongoing fees based on their adjusted gross revenue and number of urban subscribers (rural subscriber count is exempted as the goal is to bridge the digital divide). These rates are subject to periodic review.

While some of the key players in the telecom industry who invested heavily in 4G and 5G auctions have raised objections saying that it is unfair to exempt satellite communications from auctions, the TRAI has made it clear that satellites and mobile towers cannot be compared as satellite capacity is limited, the

spectrum is shared, and the technology is less flexible. Globally, many countries allocate satellite spectrum through administrative means. Additionally, faster satellite spectrum allocation brings the nation closer to its goal of bridging the connectivity gap in remote areas where traditional ground-based infrastructure is either not feasible or cost-effective.

As we eagerly await the arrival of satellite internet services in the Indian market, allow us to bring you up to speed on this trend.

All eyes are on LEO, but GEO and MEO have not lost their charm

The three major types of satellites are geostationary Earth orbit (GEO), medium Earth orbit (MEO), and low Earth orbit (LEO). As is evident from the name, LEO is closest to Earth, followed by MEO, and then GEO, which is farthest away. GEO was the first type of satellite to be used commercially, and soon thereafter came MEO. While technically, the concept of LEO can be traced back to Sputnik-1, the first